



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 12	Team No- 01	Project Title- Hostel & Mess Management System
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1. Abstract

Traditional hostel administration in educational institutions relies heavily on manual registers and spreadsheets, leading to data redundancy, inaccurate mess bill calculations, food wastage, and poor communication between students and wardens. This project proposes a polyglot, microservices-based backend system to digitize and automate the entire hostel ecosystem.

The system will evolve from a modular monolith into independently deployable microservices. A FastAPI gateway will handle authentication (JWT) and route requests to dedicated services: a Node.js/Express service for flexible user profiles and real-time activity logging (MongoDB), and a Spring Boot service for strict inventory (rooms) and transactional order (mess billing) management (PostgreSQL). The design adheres to normalized relational schemas for financial integrity and document-based schemas for user flexibility. The entire stack will be containerized using Docker, orchestrated with Kubernetes for self-healing, and integrated with a CI/CD pipeline (GitHub Actions). By providing role-based dashboards (Students, Wardens, Admins), automated attendance tracking, and real-time headcounts to reduce food wastage, the platform transforms messy manual operations into a transparent, scalable, and production-ready distributed system.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090206	K.Laxmi Lahari	
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3. Frameworks / Platforms / Software Used

FastAPI – API Gateway, JWT Auth, Pydantic validation

Node.js / Express – User & real-time notification service

Spring Boot (Java) – Inventory & billing microservice

PostgreSQL / MySQL – Relational databases for structured data

MongoDB – NoSQL database for flexible user profiles

Docker & Kubernetes – Containerization and orchestration

GitHub Actions – CI/CD automation

Guide Name and Signature (remarks if any) _____